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$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(1 + \frac{a}{bx}\right)^{cx} \\ \text{stel: } \frac{a}{bx} &= \frac{1}{u} \text{ en dus: } x = \frac{au}{b} \\ &= \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{\frac{acu}{b}} \\ &= \lim_{x \rightarrow \infty} \left(\left(1 + \frac{1}{x}\right)^u\right)^{\frac{ac}{b}} \\ &= e^{\frac{ac}{b}} \end{aligned}$$

References